Lab experiment

Tic Tac Toe Game

Coding:

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# Tic Tac Toe Game using Minimax Algorithm

board = [' ' for _ in range(9)]

# Display the board
def display():
    print()
    for i in range(0, 9, 3):
        print(board[i], "|", board[i+1], "|", board[i+2])
        if i < 6:
            print("---------")
    print()

# Check winner
def check_winner(b, player):
    win_positions = [
        (0, 1, 2), (3, 4, 5), (6, 7, 8),
        (0, 3, 6), (1, 4, 7), (2, 5, 8),
        (0, 4, 8), (2, 4, 6)
    ]

    for pos in win_positions:
        if all(b[i] == player for i in pos):
            return True

    return False

# Minimax algorithm
def minimax(b, is_max):
    if check_winner(b, 'O'):
        return 1

    if check_winner(b, 'X'):
        return -1

    if ' ' not in b:
        return 0
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    if is_max:
        best = -100

        for i in range(9):
            if b[i] == ' ':
                b[i] = 'O'
                score = minimax(b, False)
                b[i] = ' '
                best = max(best, score)

        return best

    else:
        best = 100

        for i in range(9):
            if b[i] == ' ':
                b[i] = 'X'
                score = minimax(b, True)
                b[i] = ' '
                best = min(best, score)

        return best

# Computer move
def computer_move():
    best_score = -100
    move = -1

    for i in range(9):
        if board[i] == ' ':
            board[i] = 'O'
            score = minimax(board, False)
            board[i] = ' '

            if score > best_score:
                best_score = score
                move = i

    if move != -1:
        board[move] = 'O'

# Main game
```

```
def play():
    print("Tic Tac Toe Game")
    print("You are X, Computer is O")

    while True:
        display()

        if check_winner(board, 'O'):
            print("Computer Wins!")
            break

        if ' ' not in board:
            print("Game Draw!")
            break

        try:
            pos = int(input("Enter position (1-9): ")) - 1
        except ValueError:
            print("Enter a valid number!")
            continue

        if pos < 0 or pos > 8 or board[pos] != ' ':
            print("Invalid position!")
            continue

        board[pos] = 'X'

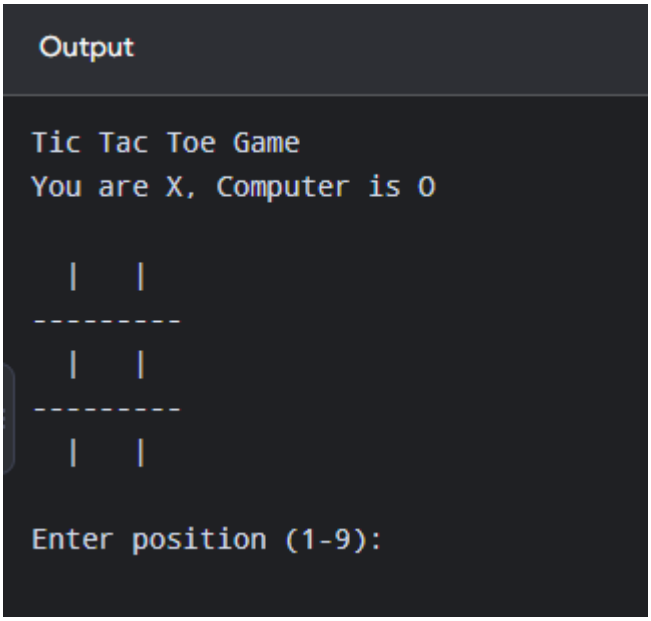
        if check_winner(board, 'X'):
            display()
            print("You Win!")
            break

        if ' ' not in board:
            display()
            print("Game Draw!")
            break

        computer_move()

play()
```

Output:


Output
Tic Tac Toe Game
You are X, Computer is O
| |

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| |
Enter position (1-9):